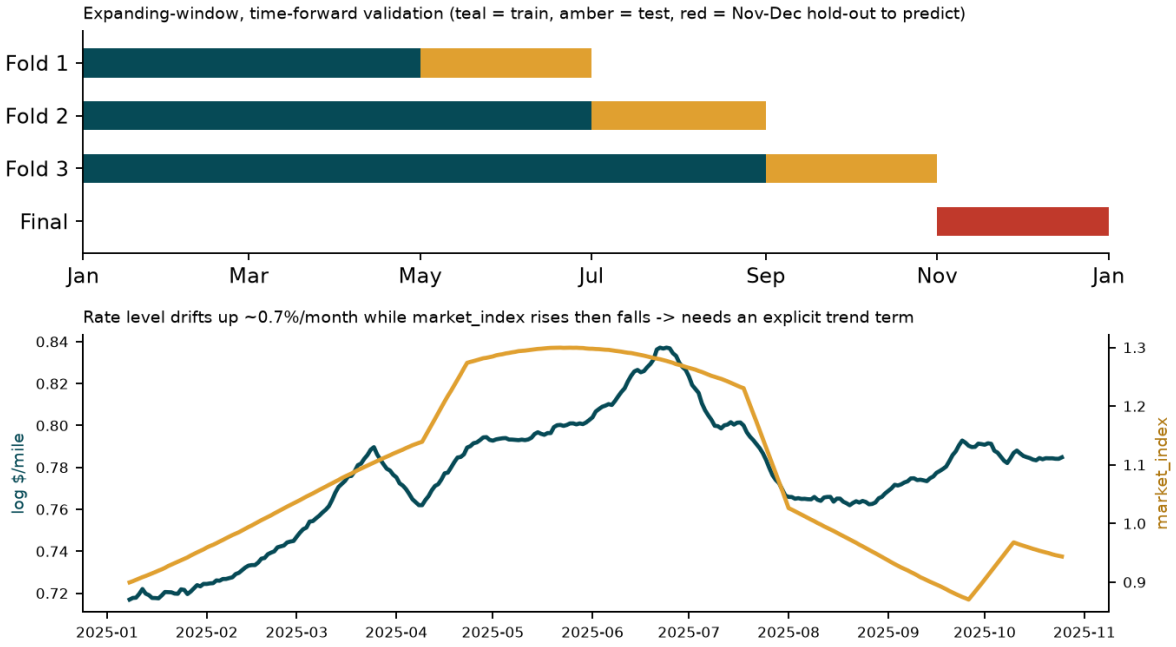
Freight Rate Prediction - Validation Approach & December Forecast

Machine Learning Engineer assessment. Model: hybrid robust trend + LightGBM residual, fitted on train_test.csv (48,000 loads, Jan-Oct 2025), applied to validation.csv (12,000 loads, Nov-Dec 2025).

1. How the data was split and why

The deliverable is a forecast of the **future**: train_test.csv covers 2025-01-01 to 2025-10-31 and validation.csv covers 2025-11-01 to 2025-12-31, so nothing in the final test overlaps in time with training. A random K-fold split would leak same-day and neighbouring-day market conditions into validation and overstate performance, so I did not use one. Instead:

- **Expanding-window, time-forward folds** with 2-month test blocks (the same length as the real hold-out): train Jan-Apr / test May-Jun; train Jan-Jun / test Jul-Aug; train Jan-Aug / test Sep-Oct. All model choices (objective, trend term, features) were made on these folds; the final model is refit on all ten months.
- **Unseen-city stress test.** 8 of the 72 cities in validation never appear in train_test.csv (about 12% of validation loads touch one). I repeatedly held out 8 random cities from training, then scored them on Sep-Oct. Error on unseen-city loads was 1.94% vs 1.86% on seen-city loads, so the coordinate-based features generalise to new cities.
- **Noisy labels handled in evaluation.** About 1.3% of labels look corrupt (0.7% with rate per mile above $5, typically 3-4x the norm, and 0.7% below $1) and they dominate RMSE/R^2. I report metrics on all rows *and* on credible rows, so model quality is not hidden by label noise nobody can predict.



2. Data-quality issues found and how they were handled

Issue (train / validation)	Handling
Negative weights (292 / 145 rows); magnitude matches valid loads	Treated as sign errors: abs(weight), plus a flag feature
Weight hard-capped at 47,500 lb (1,191 / 297 rows)	Censored values kept, flagged with weight_capped
Missing weight (300 / 165)	Left as NaN (LightGBM native) + flag; stage-1 fill with median

Issue (train / validation)	Handling
Missing market_index (374 / 249)	Filled with that day's median across loads (daily signal, in-day sd ~0.025); flag kept
Label outliers: $/mile > 5 or < 1 (646 rows, 1.3%)	Excluded from fitting only; L1 loss in the boosted stage for robustness
Distance vs. coordinates	Verified consistent (distance / haversine ~1.19, per-route jitter about 2%); no fix needed
Unseen cities in validation; train/val date shift	No city one-hot / target encoding; use coordinates, distance, equipment; explicit time trend

3. Model and key results

Key finding: log(rate per mile) depends on distance, equipment, market_index and quote_signal, but also on a **steady upward drift (~0.7% per month)** that market_index does not explain. Tree models cannot extrapolate it: a plain LightGBM holds rates at the last level it has seen, which matters because Nov-Dec is entirely outside the training dates. The chosen model is therefore two-stage: (1) a Huber-robust linear model on log $/mile with a linear time term, log market_index, quote_signal, distance terms and equipment; (2) LightGBM with an L1 objective on stage-1 residuals using route geometry, equipment, weight, weekday and data-quality flags, with no calendar-level features. Prediction = exp(stage 1 + stage 2) x distance.

Model (mean of 3 folds)	MAE $	MAPE %	WAPE %	R2	MAE $ (clean)	MAPE % (clean)
Naive $/mile by equipment	219	10.53	9.11	0.810	169	8.37
Plain LightGBM (L1)	124	5.15	5.15	0.827	73	2.86
Hybrid (chosen)	110	4.66	4.58	0.830	59	2.34

Hybrid per fold	MAE $	MAPE %	R2	MAPE % (clean)	Bias (log, clean)
Jan-Apr -> May-Jun	92	3.84	0.836	1.60	-0.005
Jan-Jun -> Jul-Aug	139	5.82	0.825	3.54	-0.033
Jan-Aug -> Sep-Oct	100	4.32	0.828	1.87	+0.014

R^2 is capped near 0.83 on all rows by the corrupt labels; on credible rows the hybrid's typical error is about 2.3%. The weakest fold (Jul-Aug, -3% bias) coincides with the market_index peak; a possible cause (not tested) is a nonlinear pass-through of the index into rates, which would be the first thing to investigate next.

4. Fixed December 2025 prediction chart (produced by score.py)

Chart inputs contain only the date, but the model also needs the two daily market features. I used the December **daily median market_index taken from validation.csv input features** (no labels) and the December median quote_signal (load-level noise with no daily structure). The result is the expected weekly cycle (midweek peak, weekend dip) on the upward trend, about $2.33/mile for a 360-mile Dry Van load ($828-$851).

